

Formal response to the v2 $\mathcal{ALCT}_{\text{RCC5}}$ updates

Review memorandum

May 2026

Abstract

This note reviews the latest v2 documents, in particular the completeness-extraction v2 paper and the sibling-interface v2 paper, with brief comments on the overview paper. The main conclusion is positive but conditional. The v2 revision correctly removes the earlier density/Hasse-cover problem by introducing witness-generated submodels and generated cover edges. It also moves the proof much closer to the intended split-forest architecture. However, several proof obligations remain open before the decidability theorem should be regarded as established. The main remaining issues concern the separation of blocking from semantic equality, the treatment of multiple incomparable upward PP-witnesses, explicit handling of EQ-modalities, finite mosaic arity, placement of DR/PO witnesses, and simultaneous realization of transitive eventualities.

1 Scope and verdict

This response uses line numbers from the supplied TeX files where useful. The main references are the completeness-extraction v2 source and the sibling-interface v2 source.

The v2 update is a substantial improvement over the earlier draft. In particular, the problematic reading of the split tree as a Hasse diagram of an arbitrary ambient PP/PPI-order has been replaced by the correct generated model picture. The witness-generated submodel lemma, generated cover edges, self-loops as blocking cycles, and support-closed mosaics are all steps in the right direction.

Nevertheless, I would not yet state the main theorem as a completed proof. The current version should be described as a strong proof programme with several remaining formal proof obligations. The most serious of these is the interaction between weak-EQ splitting, finite blocking, and multiple upward PP-witness requirements.

2 What is now fixed

2.1 The density/Hasse issue is essentially resolved

The completeness extraction now starts with witness-generated submodels. The closure definition is closed under NNF complements (completeness v2, lines 228–235), and the witness-generated submodel lemma is stated and proved at lines 314–351. The proof correctly handles the converse direction for universals by retaining a witness for $\exists R.\bar{F}$ when $\forall R.F$ fails; see lines 344–350.

This is the right bridge between ordinary abstract satisfiability and the sparse model class originally intended for the split-forest construction. It shows that arbitrary dense or non-Hasse features of a starting semantic frame are not relevant to satisfiability of the closure formulas. One may restrict to a countable submodel generated by selected witnesses.

The document also now distinguishes generated cover edges from semantic Hasse covers. Lines 405–419 explicitly say that the generated cover relation is not assumed to be a Hasse relation in the

original ambient order. This is the right conceptual correction. The proof no longer needs to read off a transitive reduction of an arbitrary PP-order.

2.2 The ambient-root assumption has been removed

The sibling-interface v2 document explicitly replaces the old ambient-root requirement. It says that v1 Definition 1.5 is reinterpreted so that there is no ambient root, generated cover edges are selected witness covers, and parent self-loops are permitted; see sibling-interface v2, lines 167–178 and 257–289.

Likewise, the completeness extraction allows self-loops in the parent function:

$$\text{par}_Q(\sigma) = \sigma.$$

This is described at lines 592–607 of the completeness file. The intended meaning is correct: a quotient self-loop represents infinitely many fresh unfolding occurrences with the same finite profile. It is not a semantic cycle and not an EQ-identification.

This handles the earlier stress formula

$$C_{\uparrow} = \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$$

under the containment orientation: each occurrence has a strict later superpart/ancestor, while the finite quotient uses a single blocking state.

2.3 The raw pair-table problem is addressed in the right direction

The v2 extraction replaces raw pair tables by support-closed mosaics; see lines 616–720 of the completeness file. This is the correct response to the witness conflation problem in the old C5-style reasoning. A mosaic records a joint atomic network on one finite vertex set, rather than aggregating independent pairwise relations that may not be jointly realized.

This is a real improvement. It preserves the original sibling-interface idea while strengthening it enough to have a plausible completeness theorem.

3 Remaining blockers

3.1 Blocking and semantic EQ are still not cleanly separated

This is the most important remaining issue.

The profile definition records a typed-EQ colour as a finite component of a rank profile; see lines 525–545. The finite-index lemma then says these colours are drawn from a finite palette; see lines 551–566. But the extracted congruence proof says that the colour is computed from the origin, choosing one colour per EQ-class in the witness-generated model; see lines 864–867. Under strong EQ, every distinct semantic element is its own EQ-class. A witness-generated model may be countably infinite, so one colour per semantic EQ-class is not a finite profile component.

There is also a direct conflict with blocking loops. Lines 592–607 correctly state that a quotient self-loop unfolds to fresh semantic occurrences. Lines 845–855 also say that repeated occurrences of the same profile are not EQ-related. However, the mosaic definition says that endpoints with the same typed-EQ colour have label EQ; see lines 693–694. If the same finite profile and colour are reused on successive laps of a blocking loop, this would incorrectly collapse the infinite PP-chain into semantic equality.

Proof obligation 1 (Separate blocking from equality). *The proof must define two distinct relations:*

$\equiv_{\text{blk}}$ *finite blocking/profile repetition,*

$\equiv_{\text{EQ}}$ *semantic equality / weak-EQ mate relation.*

Equality of finite profiles, or equality of a reusable profile colour, must never by itself imply $\equiv_{\text{EQ}}$.

Recommended repair 1 (Revise typed-EQ colours). *Do not say that a typed-EQ colour is one colour per semantic EQ-class of $\mathcal{I}_w$. Instead, make equality data explicit and local to the finite certificate. For example:*

- *The quotient contains an explicit finite relation $\equiv_{\text{EQ},Q}$ on named equality ports, core copies, or mate slots.*
- *Blocking states may repeat the same profile, but every unfolding lap gets fresh equality ports unless the certificate explicitly links them by $\equiv_{\text{EQ},Q}$.*
- *The mosaic condition should be changed from “same colour implies EQ” to “EQ-label iff the endpoints are identified by the explicit equality relation visible in that mosaic.”*
- *Add a prohibition that no two occurrences connected by a strict vertical PP/PPI-path may be EQ-identified.*

With this change, blocking is purely a finite representation device, while strong EQ is handled only by the explicit typed congruence.

3.2 Multiple upward PP witnesses need an EQ-aware construction

The containment orientation uses parent edges for superparts. Therefore $\exists \text{PP}.D$ requirements point upward. A single occurrence in a tree has only one parent chain. But a semantic element may need several incomparable proper-superpart witnesses.

A useful stress formula is

$$\begin{aligned} C_{AB} := & \exists \text{PP}.A \sqcap \exists \text{PP}.B \\ & \sqcap \forall \text{PP}(\neg A \sqcup (\forall \text{PP}.\neg B \sqcap \forall \text{PPI}.\neg B)) \\ & \sqcap \forall \text{PP}(\neg B \sqcup (\forall \text{PP}.\neg A \sqcap \forall \text{PPI}.\neg A)). \end{aligned}$$

This concept is satisfiable by an element x with two incomparable proper superparts y and z , where $y \models A$, $z \models B$, and $y \text{PO} z$. The universal conjuncts prevent the A - and B -witnesses from being comparable along one parent chain.

Thus a single occurrence of x cannot realize both PP-witnesses by one linear parent chain. The original split-copy idea is designed exactly for this case: create weak-EQ copies of x , put one copy under the A -superpart and another copy under the B -superpart, and then recover strong EQ by a typed congruence.

The current v2 reachability clause is still too local. It says, essentially,

$$\exists R.D \in \lambda_Q(\sigma) \implies \exists \sigma' (\sigma \rightsquigarrow_R \sigma' \text{ and } D \in \lambda_Q(\sigma'))$$

for $R \in \{\text{PP}, \text{PPI}\}$; see lines 960–979 and 1067–1075. This does not say that a witness may be realized along the parent chain of an EQ-mate of σ .

Proof obligation 2 (EQ-aware reachability). *For split copies, existential and universal PP/PPI clauses must be checked on equality classes, not on a single occurrence profile. A suitable existential condition is:*

$$\exists \text{PP}.D \in \lambda(\sigma) \Rightarrow \exists \hat{\sigma} \in \text{Mate}(\sigma) \exists \sigma' (\hat{\sigma} \rightsquigarrow_{\text{PP}} \sigma' \wedge D \in \lambda(\sigma')),$$

and similarly for PPI. A suitable universal condition is:

$$\forall \text{PP}.D \in \lambda(\sigma) \Rightarrow \forall \hat{\sigma} \in \text{Mate}(\sigma) \forall \sigma' (\hat{\sigma} \rightsquigarrow_{\text{PP}} \sigma' \Rightarrow D \in \lambda(\sigma')).$$

Here $\text{Mate}(\sigma)$ must be derived from explicit typed-EQ mate data, not from profile equality.

This repair is needed for completeness. After quotienting the weak split model by $\equiv_{\text{EQ}}$, a witness found above any mate copy becomes a witness for the strong semantic element.

3.3 The role EQ must be normalized or handled explicitly

The language treats the five RCC5 atoms as modal roles, including EQ. The validity proof distinguishes only DR/PO witnesses and PP/PPI witnesses; see lines 1067–1075. There is no case for $R = \text{EQ}$.

Under strong equality semantics this is easy to fix. Since

$$\exists \text{EQ}.D \equiv D, \quad \forall \text{EQ}.D \equiv D,$$

all EQ-modalities may be eliminated during preprocessing, or the type conditions may explicitly enforce these equivalences.

Recommended repair 2 (EQ-normalization). *Add a preprocessing lemma: for every concept C , compute in polynomial time a concept $C^{-\text{EQ}}$ in which $\exists \text{EQ}.D$ and $\forall \text{EQ}.D$ are replaced by D , recursively, and prove equivalence under strong EQ. Then the rest of the proof may assume existential roles are only DR, PO, PP, PPI.*

Alternatively, add EQ as a third case in the validity proof.

3.4 The finite mosaic arity bound is not yet proved

The support-closed mosaic idea is sound in spirit, but the finite arity bound is not yet formal. Lines 722–733 say mosaics may be bounded by $|\text{Cl}(C_0)| + 4$ vertices because larger mosaics decompose into overlapping smaller ones. Lines 1258–1264 then acknowledge that a rigorous treatment would derive this bound from a Ramsey-type argument.

This is not just a cosmetic matter. The decidability proof depends on an effectively enumerable finite certificate class. Therefore the arity bound must be a theorem, not an informal statement.

Proof obligation 3 (Finite mosaic locality). *Prove that there is an effectively computable number $N(C_0)$ such that every finite prefix network needed by the split-forest soundness construction is controlled by support-closed mosaics of arity at most $N(C_0)$. Equivalently, show that all required restriction, triple-composition, typed-EQ replacement, containment-extension, and witness-support checks are local below that bound.*

One possible route is to define $N(C_0)$ from the number of endpoint summaries and closure-existential requirements, rather than asserting $|\text{Cl}(C_0)| + 4$. A second route is to state and prove a patchwork/locality theorem for abstract RCC5 mosaics with typed endpoint summaries.

3.5 Placement of DR/PO witnesses remains schematic

The witness-menu construction for DR/PO existentials is at lines 775–795. It says that the selected witness occurrence exists in the split tree, possibly as a sibling under a common ancestor. But the construction that places selected DR/PO witnesses into side contexts or sibling mosaics is not given.

For completeness, the proof needs to explain how every selected DR- or PO-witness from the witness-generated submodel is represented in the containment split presentation while preserving:

- the chosen DR/PO relation to the source occurrence;
- all universal restrictions at the source and witness;
- all composition constraints with the vertical ancestors and descendants;
- consistency with sibling mosaics and typed-EQ mate data.

Recommended repair 3 (Side-context witness lemma). *Add a lemma of the following form.*

Lemma. Let u be an occurrence and let $b = w(\text{orig}(u), \exists R.D)$ with $R \in \{\text{DR}, \text{PO}\}$. Then the generated split presentation can be extended, by adding a bounded side context and a support-closed mosaic, with an occurrence v of origin b such that $\rho(u, v) = R$, $D \in \lambda(v)$, and all relation-safety and RCC5 composition constraints involving the finite interface are preserved.

Without such a lemma, the witness-menu entries are not yet backed by a formal model-to-certificate construction.

3.6 Reachability discharge must be simultaneous and coherent

The reachability formulation is much cleaner than the earlier automata-heavy presentation. In a deterministic parent graph, a PP-eventuality can often be reduced to reachability along the unique parent walk. However, the paper itself notes that simultaneous discharge requires a single coherent infinite walk; see lines 1266–1274.

The current argument says this follows from finiteness and a patchwork property. That needs to be a proof, especially once split copies and EQ-aware reachability are introduced. The issue is not just whether each eventuality has some reachable witness profile, but whether all eventualities required along one unfolded branch can be realized together in the same unfolding.

Proof obligation 4 (Coherent eventuality realization). *Add a finite-cycle realization lemma. A possible statement is:*

Lemma. Let G_Q be the finite parent graph equipped with explicit equality-mate data. If every PP/PPI-eventuality satisfies the EQ-aware reachability condition and every universal is closed under the relevant reachability relation, then the canonical unfolding of G_Q realizes all vertical eventualities simultaneously.

For PP under a functional parent map, this can be proved by analyzing the unique ultimately periodic parent walk. For PPI, where inverse-parent branching may be non-deterministic, the proof must use the child multiplicity component of the profile to show that each occurrence has the required child branch witnesses.

3.7 Minor source issue: undefined

`\sst`

The completeness source uses `\sst` in the status partition at lines 632–638, but no macro `\sst` appears to be defined. This causes an undefined-control-sequence error under ordinary LaTeX compilation. The source should either define

$$\newcommand{\sst}{\operatorname{st}}$$

or replace `\sst` with the already defined status notation.

4 Assessment of the companion sibling-interface v2 paper

The sibling-interface v2 note is useful and should be kept as a delta document. It correctly states that the old soundness chain is retained while the ambient root and v1 decidability sketch are replaced. In particular, the explicit list of retained and replaced material at lines 131–180 is helpful.

However, the sibling-interface note inherits the same unresolved obligations from the completeness paper. In particular:

1. Self-loops must be blocking only, never equality. The note says this at lines 409–414, but the full finite certificate still needs explicit equality data that cannot identify blocking laps.
2. Reachability discharge at lines 459–470 must be made EQ-aware to handle split copies with several incomparable upward PP-witnesses.
3. The theorem at lines 534–580 relies on the companion completeness theorem; therefore it should be stated conditionally until the obligations in this response are closed.

A good wording would be:

Together with a complete proof of the v2 completeness-extraction theorem, the preserved v1 soundness chain yields decidability.

Rather than:

Combined with v2 completeness extraction, this establishes decidability.

The latter is too strong until the completeness extraction is fully formalized.

5 Comments on the overview document

The overview contains a valuable disclaimer saying that the results have not been peer reviewed and should not be taken as established without independent verification. That disclaimer should remain.

The section describing the theoretical layer should be softened. The current language says that the theoretical layer establishes decidability and that the two theoretical papers establish satisfiability iff existence of a valid finite rank- d quotient. Given the remaining obligations above, a safer statement is:

The theoretical layer proposes a route to decidability by reducing satisfiability to the existence of a valid finite generated split-forest quotient. The current v2 manuscripts address the earlier density/Hasse and raw pair-table defects, but the completeness extraction still depends on explicit proofs of typed-EQ separation, EQ-aware reachability, finite mosaic locality, and coherent eventuality realization.

6 Suggested priority order for revision

The following order would make the next revision maximally efficient.

1. **Fix equality data first.** Separate $\equiv_{\text{blk}}$ from $\equiv_{\text{EQ}}$, remove any implication from profile equality or reusable colour equality to semantic EQ, and make typed equality an explicit finite relation in the certificate.
2. **Make vertical reachability equality-class aware.** Add mate-based existential and universal reachability clauses for PP/PPI.
3. **Normalize EQ-modalities.** Either eliminate them by a preprocessing lemma or add a direct validity case.
4. **Prove finite mosaic locality.** Replace the informal $|\text{Cl}(C_0)| + 4$ claim by a theorem with an effective bound.
5. **Add the side-context lemma for DR/PO witnesses.**
6. **Prove simultaneous eventuality realization.** This may be a direct lasso/branch proof rather than a Buechi automaton construction.
7. **Repair the TeX source.** Define or remove `\sst` and ensure clean compilation without nonstop-mode errors.

7 Conclusion

The v2 updates fix the most visible structural problem in the earlier proof: the proof no longer depends on a Hasse diagram or ambient root of an arbitrary semantic PP/PPI-order. The witness-generated submodel lemma and generated cover reinterpretation put the argument much closer to the intended model class. The support-closed mosaic idea also repairs the earlier raw pair-table weakness in the right conceptual direction.

The decidability proof is therefore significantly more plausible than before. However, it is not yet complete as a formal proof. The key missing piece is a fully specified finite certificate in which blocking, split-copy equality, vertical reachability, and sibling mosaics interact coherently. Once the remaining obligations listed above are discharged, the original split-forest strategy may well yield the desired decidability theorem.